

SAN PEDRO SULA MORALES, Honduras

WMO: 787080

Lat: 15.453N

Lon: 87.924W

Elev: 91

StdP: 14.65

Time Zone: -6.00 (W06)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	64.1	65.8	60.4	78.7	74.2	62.4	84.7	73.4	17.6	80.9	15.8	81.6	2.8	210	0.298

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	18.2	98.3	78.6	96.0	79.0	94.7	79.1	83.7	90.8	82.8	89.5	82.1	88.6	9.3	40

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
82.3	168.6	87.2	81.0	161.5	85.8	80.7	159.8	85.5	47.8	91.5	46.9	89.5	46.1	89.2	87.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
18.4	16.4	14.4	DB	60.8	103.9	1.9	2.7	59.4	105.9	58.3	107.5	57.2	109.1	55.8	111.0
			WB	59.6	85.7	2.1	1.4	58.0	86.7	56.8	87.5	55.6	88.2	54.0	89.2

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	80.9	75.6	77.9	79.7	83.4	84.7	84.0	82.9	83.5	83.9	81.3	77.5	76.4	(6)
	DBStd	4.47	3.74	3.78	4.34	3.82	2.68	2.14	1.69	1.73	1.84	3.16	3.71	3.59	(7)
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0	(9)
	CDD50	11284	793	780	922	1001	1076	1020	1021	1038	1017	971	826	819	(10)
	CDD65	5809	328	360	457	551	611	570	556	573	567	506	376	354	(11)
	CDH74	55560	1994	2767	4136	6023	7010	6277	5799	6112	6123	4581	2516	2222	(12)
	CDH80	23867	646	1075	1822	2887	3319	2867	2477	2739	2790	1761	792	692	(13)
(14) Wind	WSAvg	5.2	4.4	5.0	6.0	6.3	6.4	5.5	5.5	5.4	5.3	4.4	4.0	4.2	(14)
(15) Precipitation	PrecAvg	46.9	2.8	2.2	1.5	1.4	2.9	6.6	3.7	4.1	6.4	6.0	5.8	4.2	(15)
	PrecMax	64.2	10.5	6.8	4.5	5.9	11.8	14.0	9.9	8.5	21.6	17.0	15.1	9.7	(16)
	PrecMin	20.7	0.2	0.0	0.0	0.0	0.3	0.5	0.1	0.6	1.0	1.1	0.6	0.6	(17)
	PrecStd	11.9	2.0	1.6	1.3	1.5	2.6	3.5	1.9	1.8	4.1	3.6	3.2	2.4	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	91.8	95.1	100.3	102.5	100.1	97.2	95.1	96.4	97.2	95.3	93.1	91.8	(19)
		MCWB	76.0	75.9	75.1	78.5	79.4	79.4	79.4	79.9	78.6	79.6	77.7	77.3	(20)
	2%	DB	88.4	91.6	95.3	98.4	96.9	95.2	93.2	94.4	95.1	93.2	89.7	88.6	(21)
		MCWB	76.5	76.2	76.6	78.9	79.9	79.7	79.3	79.8	79.4	79.8	78.6	77.7	(22)
	5%	DB	86.1	89.3	92.5	95.0	95.0	93.4	91.6	92.9	93.3	91.2	87.6	86.2	(23)
		MCWB	76.2	76.5	77.0	79.1	79.6	79.6	79.2	79.8	79.8	79.5	78.3	77.3	(24)
	10%	DB	84.0	86.4	89.5	92.3	93.1	91.5	90.0	91.3	91.6	89.2	84.6	84.1	(25)
		MCWB	75.7	76.1	76.5	78.7	79.3	79.5	79.3	79.6	79.6	79.1	77.7	76.6	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	79.8	80.6	82.5	84.5	84.5	84.4	83.6	83.9	84.2	83.4	82.0	81.5	(27)
		MCDB	85.7	88.5	90.6	92.8	93.2	91.4	89.6	90.7	91.3	89.9	86.8	86.9	(28)
	2%	WB	78.5	79.0	80.1	82.5	82.8	82.9	82.4	82.7	82.8	82.0	80.6	79.8	(29)
		MCDB	84.3	86.2	88.0	90.4	90.4	89.3	88.1	89.0	89.5	88.1	86.0	84.8	(30)
	5%	WB	77.4	78.0	78.9	81.1	81.7	81.9	81.5	81.7	81.9	81.0	79.4	78.6	(31)
		MCDB	83.3	84.8	86.3	89.0	88.9	88.1	87.3	88.0	88.4	87.1	84.2	83.4	(32)
	10%	WB	76.3	77.0	77.8	80.0	80.8	81.0	80.7	80.9	81.0	80.0	78.4	77.4	(33)
		MCDB	81.9	83.4	84.9	87.6	87.9	87.2	86.5	87.3	87.5	85.8	82.8	81.9	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	14.7	16.8	18.9	19.6	18.2	17.2	16.6	17.4	17.8	14.8	13.3	14.2	(35)
		MCDBR	19.1	21.1	23.6	23.1	20.5	18.8	17.9	18.9	19.6	18.3	17.2	17.9	(36)
	5% WB	MCWBR	8.9	9.0	8.8	8.2	7.1	7.2	6.7	7.2	7.6	7.1	7.5	8.4	(37)
		MCWBR	17.2	18.9	20.4	20.7	18.8	17.2	16.4	17.4	17.6	16.1	14.7	15.7	(38)
(40) Clear-Sky Solar Irradiance	taub	taub	0.385	0.401	0.434	0.527	0.577	0.486	0.488	0.473	0.455	0.427	0.406	0.413	(40)
		taud	2.506	2.458	2.352	2.065	1.945	2.245	2.222	2.276	2.324	2.425	2.469	2.431	(41)
	Ebn at Noon	Ebn at Noon	288	288	280	253	237	257	257	264	270	276	279	275	(42)
		Edh at Noon	33	36	41	55	60	44	45	43	41	36	34	35	(43)
(44) All-Sky Solar Radiation	RadAvg	RadAvg	1178	1504	1729	1807	1694	1688	1733	1756	1680	1327	1136	1127	(44)
	RadStd	RadStd	148	138	133	80	106	108	69	78	70	155	107	116	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (5 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page